HADOOP HOMOGENEOUS MULTINODE CLUSTER - FINAL GUIDE

ARCHITECTURE:
1 Master + 3 Slaves (Slave3 runs SecondaryNameNode)

STEP 1: /etc/hosts (ALL SYSTEMS)
192.168.1.100 master
192.168.1.101 slave1
192.168.1.102 slave2
192.168.1.103 slave3

STEP 2: CREATE USER
sudo adduser hadoop
sudo usermod -aG sudo hadoop
su - hadoop

STEP 3: INSTALL JAVA
sudo apt install openjdk-11-jdk

STEP 4: INSTALL HADOOP & SET ENV
export HADOOP_HOME=~/hadoop
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin

Edit hadoop-env.sh:
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64

STEP 5: PASSWORDLESS SSH (MASTER)
ssh-keygen -t rsa
ssh-copy-id slave1
ssh-copy-id slave2
ssh-copy-id slave3

STEP 6: MASTER CONFIG
core-site.xml -> fs.defaultFS = hdfs://master:9000
hdfs-site.xml -> dfs.replication=3
dfs.namenode.name.dir=/home/hadoop/hadoopdata/namenode
mapred-site.xml -> mapreduce.framework.name=yarn
yarn-site.xml -> yarn.resourcemanager.hostname=master
workers -> slave1 slave2 slave3

STEP 7: COPY CONFIG TO SLAVES
scp -r $HADOOP_HOME/etc/hadoop slave1:$HADOOP_HOME/etc/

STEP 8: SLAVE hdfs-site.xml
dfs.datanode.data.dir=/home/hadoop/hadoopdata/datanode

STEP 9: CREATE DIRECTORIES
mkdir -p ~/hadoopdata/namenode
mkdir -p ~/hadoopdata/datanode

STEP 10: FORMAT & START
hdfs namenode -format

start-dfs.sh
start-yarn.sh

SLAVE3:
hdfs secondarynamenode &

VERIFY:
jps